

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

27th July 2022

**Subject CP1 – Actuarial Practice
(Paper B)**

Time allowed: 3 Hours 30 Minutes (09.30 - 13.00 Hours)

Total Marks: 100

Q. 1) A large local manufacturing company currently sponsors a defined benefit pension scheme for its employees. The scheme provides annual pension equal to 1/60th of final basic salary for each year of accrued service post retirement. The employees also have an option to commute the entire benefit payable to lumpsum in cash on retirement rather than opting to receive pension for life.

- i)** Briefly describe the funded approaches available to providers of pension to cover the liabilities of the pension scheme. (10)

The manufacturing company has closed the DB scheme to new joiners and transferred the asset and liabilities of the scheme to an established insurer in the country.

- ii)** What are the implications to the manufacturing company because of transfer of scheme to the insurer in comparison with continuing to self-manage the scheme? (6)

At the time of transfer, the Valuation Actuary employed by the insurer has recommended the use of the raw mortality rates derived from the past experience of the scheme in order to set the mortality assumptions to determine the transfer value.

- iii)** Highlight the reasons why it may not be a suitable recommendation? (7)

The Highway Authority of the country has recently completed toll road project which covers the distance between two economically active cities 200 kms apart. The Authority wants to transfer the legal rights to operate the toll road for a fixed pre-determined period to a private operator. The private operator will get regular cash flow in the form of collection of tolls for a fixed period of 25 years in return of initial lumpsum payment to the Highway Authority. The tolls will be decided by the Authority from time to time. Post completion of 25-year period, the project will be transferred back, without any charge to the Highway Authority.

- iv)** Why would Highway Authority want to transfer the project to a private operator? (4)

Based on internal analysis done by the investment department of the insurer, the investment in the toll road project is expected to provide a high rate of return post allowing for expected operational and maintenance costs. Therefore, the Chief Investment Officer has suggested to invest part of the fund received as part of the DB scheme transfer in the toll road project.

- v)** Outline the risks that the insurer is exposed to due to the proposed investment in the toll road project. (13)

The insurer has recently performed the analysis of its pension portfolio as part of the regular review. The insurer has observed that the proportion of members opting to take lumpsum in cash for the above scheme are found to be much higher than other schemes with similar benefit structure in the insurer's portfolio.

- vi)** What are the likely reasons for higher proportion of members taking the commutation option from this scheme i.e. take lumpsum in cash? (10)
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Q. 2) An insurance company is writing business in health and life insurance. The Board of Directors received the following proposal from the management team:

Proposal 1:

All the existing and prospective health insurance customers will be given a choice to enrol in a wellness scheme. The wellness scheme mainly involves the tracking of health related information like calories burnt based on steps covered or time spent in gym or yoga centre. This information will be collected through a wearable gadget. Based on this information, the customer will earn certain number of points using pre-defined criteria. The pre-defined criteria mainly depend upon age of the policyholder and physical activities carried out over a period of one year. The customer can redeem the points annually or can carry forward up to a maximum period of 3 years. These points can be redeemed for discount in renewal premium based on pre-defined criteria.

Proposal 2:

In order to reduce the cost of health care for old age population, a new immediate annuity product with a fixed benefit of an amount equal to one-year annuity on hospitalisation in addition to regular fixed annuity may be launched. Such a fixed benefit is payable once a year and the benefit is irrespective of cost incurred on hospitalisation.

- i) Discuss the merits and demerits of the Proposal 1 to the insurer. (5)
 - ii) Outline the risks from insurer's point of view in respect of Proposal 1. (8)
 - iii) Briefly discuss the ways to mitigate those risks. (5)
 - iv) Explain the following stages involved in deciding the pre-defined criteria for earning the points and redemption of those points by policyholders:
 - Arriving at the savings in claims cost.
 - Linking calories burnt, points earned and discount in renewal premium.
 - Redemption criteria.
 - Other costs to be considered.
 (9)
 - v) State the aims of regulation and describe the expected areas of regulation in respect of above two proposals. (7)
- The regulations stipulate that the insurer needs to hold a reserve for liability arising under Proposal 1 and Proposal 2 separately. It also stipulates that the assets supporting such reserves shall be earmarked explicitly.
- vi) State the sources of data and assumptions that need to be made for setting up these reserves. (6)
 - vii) State the principles of investment and discuss the suitability of bonds, equity and money market instruments as assets in support of these reserves. (10)

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